

# Thresher

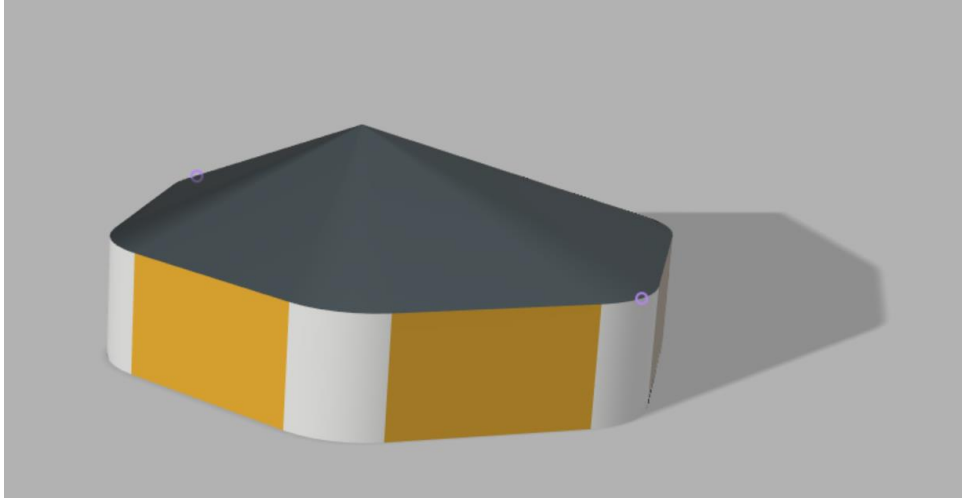
## Intro/context:

Sharks are vertebrates, like us humans, but they are considered fish. They are made from cartilage, which is what our ears and nose are made of. They do eat multiple organisms, mostly smaller fish, but they are mostly scavengers meaning that they consume dead animals. Sharks are primitive animals, having skeletons made of cartilage rather than bone. They lack scales, but have small, tooth-like plates, called denticles, embedded in their skin making it extremely abrasive. Larger sharks feed exclusively on plankton and swim slowly with their mouth open, trapping plankton in their gills. Sharks have been poached for their fins, in Asia, for soup and/or their livers (which are rich in Vitamin A) for food and sport.

## Why save Sharks when they kill human beings?

Sharks are an important part of the Food chain, but they are endangered species. Human activities such as overfishing, habitat disruption, and changes in coastal water use can increase the likelihood of shark-human encounters, but scientific evidence shows that sharks do not view humans as prey and do not intentionally hunt humans as food. By creating this tag and tracking their location and good habits, sharks can be saved. Ultimately humans will be also saved because of this project as sharks will not wander off to the coastal areas where humans are, but they will be contained in their natural habitat. Overfishing can be reduced, especially by knowing where they intentionally or unintentionally get caught. When this data is collected, it can help save their habitat and reduce the pollution of their food source. This project will help with the conservation and research. Our tag will last for approximately 25 years while acoustic tags last for only 10 years and it's much safer to add on as it will go between the dorsal fin and the flipper while the satellite tags go on the dorsal fin.

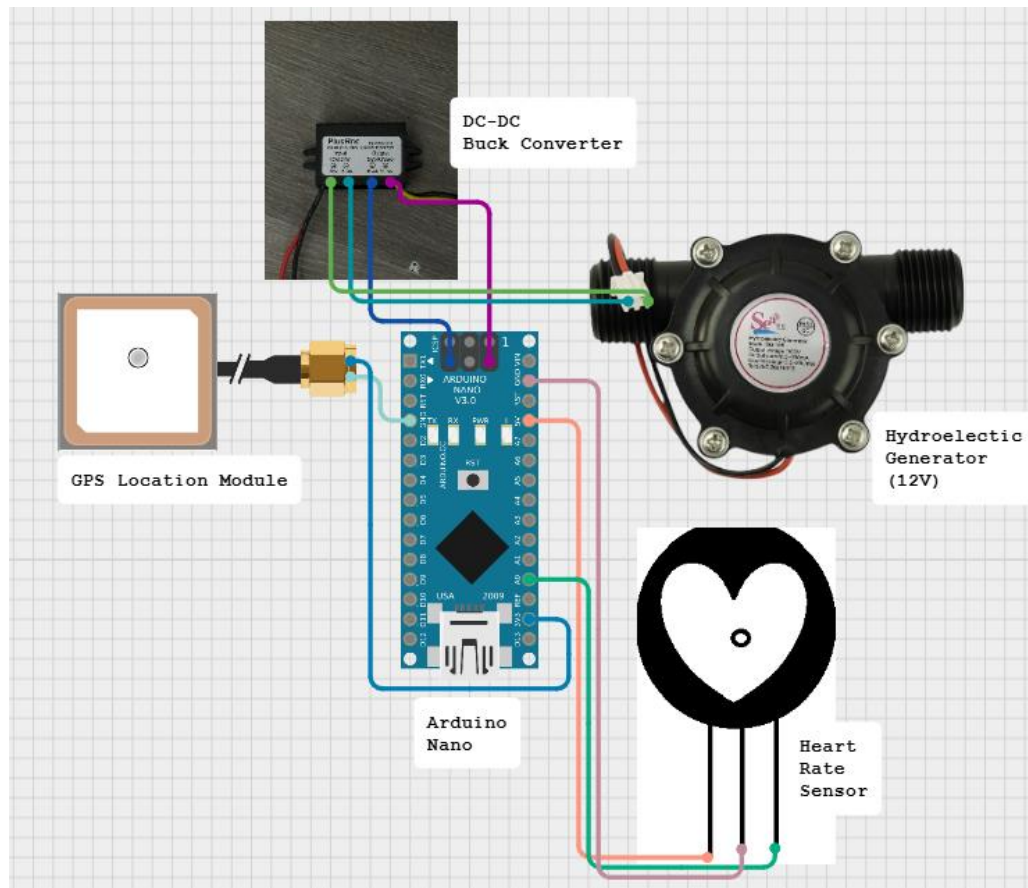
Why does the Thresher Tag look like the way it does?



The Thresher tag is made in this particular shape because it has been optimized for a firm grip on the shark, quick identification, and for aerodynamics when underwater. The Thresher tag is made of composite materials such as neoprene, rubber, and nylon, giving it a robust structure when latched onto the shark's body underwater. Additionally, a small sized wire at the bottom not only serves as a detector, but also as an addition to the existing structure of firmly gripping onto the shark's body. The bright yellow and white combination on the sides contributes to the uniqueness of the Thresher tag. It makes the tag easily identifiable to divers and researchers who study sharks. Lastly, the Thresher

tag's design is aerodynamic, making it suitable for oceanic conditions and the agility and speed of the sharks. Moreover, the hexagonal shape of the Thresher tag gives it the aerodynamic quality, which is not found in any shark tracker.

How does the Thresher tag work?



The circuit above shows the circuit for the shark tag and has a hydroelectric generator, a GPS module, a heart rate monitor, a 12V to 5V DC-DC (Buck) Converter, and a microcontroller (the Arduino Nano). This circuit will give us information on where the shark is and what the shark is eating while the shark is in the water. You may be thinking how does this help people know if the shark is eating? Well, the shark can get 'excited' if it sees its prey and that affects the heart rate, so then, we can know what it is eating. The GPS Module will help scientists understand where the shark is through satellite connectivity from underwater. The Hydroelectric generator will generate 12V for the Arduino Nano so that it can run longer than the previous tags. The buck converter will be used to convert the 12V from the hydroelectric generator to 5V for the Arduino Nano in a safe way otherwise the Arduino Nano will not be able to function properly.

The website's code (link at bottom) will be used because the formula:  $S = k (E + B + H + A) e^{-\lambda F}$  where S is the shark foraging suitability, k for scaling or normalizing constant, E for environmental influence, B for biological factors, H for habitat structure, A for anthropologic structure,  $\lambda$  for damping factor controlling sensitivity to risk, and F for foraging energy/risk field.

How will this impact the sharks and humans?

The Thresher Tag will have a positive impact on both sharks and humans. For sharks, the tag will help scientists monitor their behavior, location, and feeding patterns without harming them. This information can be used to protect their habitats, prevent overfishing, and ensure they stay within safe ocean zones. The collected data will also help reduce the number of sharks accidentally caught in fishing nets and improve conservation policies.

For humans, this means fewer shark attacks, safer coastal areas, and a better understanding of ocean ecosystems. As sharks are crucial to maintaining the balance of marine food chains, protecting them ultimately protects ocean health — and by extension, human life. By using the Thresher Tag, we can promote peaceful coexistence between humans and sharks while advancing marine science and sustainability.